Database Schema Documentation for `stockmaster3000`

The `stockmaster3000` database is designed to manage data related to users, categories, inventories, suppliers, products, and reports. This schema provides a comprehensive structure for managing stock and inventory systems while ensuring data integrity and change tracking through auditing. The application uses Hibernate to auto-generate the schema, but this SQL schema serves as a backup in case anything goes wrong. Below is the detailed documentation for the structure of this database, including tables, relationships, and data types.

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1. Users Table (`_users`)

This table stores information about users who interact with the system.

Column Name	Data Type	Description
`id`	`BIGINT`	Primary key, auto-incremented.
`username`	`VARCHAR(50)`	Unique username for each user.
`password`	`VARCHAR(255)`	User's password (hashed).
`created_at`	`TIMESTAMP`	Timestamp of when the user was created. Default is the current timestamp.
`updated_at`	`TIMESTAMP`	Timestamp of when the user was last updated. Updates automatically on changes.

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2. Categories Table (`categories`)

This table stores product categories for classification purposes.

Column Name	Data Type	Description
`id`	`BIGINT`	Primary key, auto-incremented.

`name`	`VARCHAR(255)`	Unique category name (e.g., Food, Beverages, Electronics).
`created_at`	`TIMESTAMP`	Timestamp of when the category was created. Default is the current timestamp.
`updated_at`	`TIMESTAMP`	Timestamp of when the category was last updated. Updates automatically on changes.

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3. Inventories Table (`inventories`)

This table stores inventory items such as fridges, warehouses, or stores, which can hold products.

Column Name	Data Type	Description
`id`	`BIGINT`	Primary key, auto-incremented.
`name`	`VARCHAR(255)`	Name of the inventory (e.g., Fridge).
`user_id`	`BIGINT`	Optional foreign key linking to a user who owns the inventory.
`created_at`	`TIMESTAMP`	Timestamp of when the inventory item was created. Default is the current timestamp.
`updated_at`	`TIMESTAMP`	Timestamp of when the inventory item was last updated. Updates automatically on changes.

Foreign Keys:

- `user_id` references `_users(id)`, linking inventories to users.

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4. Suppliers Table (`suppliers`)

This table stores information about product suppliers.

Column Name	Data Type	Description
`id`	`BIGINT`	Primary key, auto-incremented.

name	VARCHAR(255)	Unique supplier name.
created_at	TIMESTAMP	Timestamp of when the supplier was added. Default is the current timestamp.
updated_at	TIMESTAMP	Timestamp of when the supplier was last updated. Updates automatically on changes.

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5. Reports Table (`reports`)

This table stores reports related to inventories and products.

Column Name	Data Type	Description
id	BIGINT	Primary key, auto-incremented.
summary	TEXT	Summary of the report.
inventory_id	BIGINT	Foreign key referencing the inventory item related to the report.
created_at	TIMESTAMP	Timestamp of when the report was created. Default is the current timestamp.
updated_at	TIMESTAMP	Timestamp of when the report was last updated. Updates automatically on changes.

Foreign Keys:

- `inventory_id` references `inventories(id)`, linking the report to an inventory.

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6. Products Table (`products`)

This table stores information about products, including details such as price, quantity, and expiration.

Column Name	Data Type	Description
id	BIGINT	Primary key, auto-incremented.

`name`	`VARCHAR(255)`	Name of the product.
`price`	`DOUBLE`	Price of the product.
`quantity`	`INT`	Quantity of the product in stock.
`nutritions`	`TEXT`	Nutritional values of the product (optional).
`amountOfDaysUntilExpiration`	`INT`	Number of days until expiration (optional).
`supplier_id`	`BIGINT`	Foreign key linking to the supplier.
`category_id`	`BIGINT`	Foreign key linking to the product category.
`inventory_id`	`BIGINT`	Foreign key linking to the inventory where the product is stored.
`created_at`	`TIMESTAMP`	Timestamp of when the product was added. Default is the current timestamp.
`updated_at`	`TIMESTAMP`	Timestamp of when the product was last updated. Updates automatically on changes.

Foreign Keys:

- `supplier_id` references `suppliers(id)`, linking the product to a supplier.
- `category_id` references `categories(id)`, linking the product to a category.
- `inventory_id` references `inventories(id)`, linking the product to an inventory.

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7. User Management

The `saed` user is created with the following credentials:

- **Username:** `saed`
- **Password:** `12345678` (hashed)

This user has been granted full privileges on the `stockmaster3000` database. The privileges include:

- `SELECT`, `INSERT`, `UPDATE`, and `DELETE` permissions on all tables in the `stockmaster3000` database.

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8. Hibernate Envers and Auditing

Hibernate Envers is enabled for auditing purposes, specifically for tracking changes to the `products` table. It automatically creates an audit table (`products_aud`) that stores historical records of product changes.

Audit Table for `products` (`products_aud`)

Column Name	Data Type	Description
`id`	`BIGINT`	The primary key of the product in the `products` table.
`rev`	`BIGINT`	The revision number representing the specific version of the product record.
`revtype`	`INT`	The type of operation performed on the product record. `0` for Insert, `1` for Update, `2` for Delete.
`name`	`VARCHAR(255)`	The name of the product at the time of the revision.
`price`	`DOUBLE`	The price of the product at the time of the revision.
`quantity`	`INT`	The quantity of the product at the time of the revision.
`nutritions`	`TEXT`	Nutritional values of the product at the time of the revision (optional).
`amountOfDaysUntilExpiration`	`INT`	Number of days until expiration at the time of the revision (optional).
`supplier_id`	`BIGINT`	The supplier associated with the product at the time of the revision.
`category_id`	`BIGINT`	The category associated with the product at the time of the revision.
`inventory_id`	`BIGINT`	The inventory where the product was stored at the time of the revision.
`created_at`	`TIMESTAMP`	The timestamp when the product record was created.
`updated_at`	`TIMESTAMP`	The timestamp when the product record was last updated.

Relationships Overview

- **Users and Inventories**: A user can own multiple inventories, but each inventory is optionally linked to a single user.
- **Categories and Products**: A product is associated with a single category.
- **Inventories and Products**: A product is stored in a specific inventory.
- **Suppliers and Products**: A supplier can provide multiple products, and each product is linked to a supplier.
- **Reports and Inventories**: A report is associated with a specific inventory.

SQL Script Overview

1. **Database Creation**: Creates the `stockmaster3000` database and switches the context to it.
2. **Table Creation**: Creates six tables: `_users`, `categories`, `inventories`, `suppliers`, `reports`, and `products`.
3. **Foreign Key Constraints**: Ensures relational integrity between tables like `users` to `inventories`, `suppliers` to `products`, etc.
4. **Default Data**: Inserts initial records into the `categories` and `suppliers` tables.
5. **User Management**: Creates a user named `saed` and grants them all privileges on the `stockmaster3000` database.
6. **Audit Tables**: Hibernate Envers automatically creates audit tables to track changes to the `products` table.
